

1] Assume: Ideal gas⁺² Given: T_1, P_1, T_2, P_2 (in spreadsheet)
 $R = 287 \text{ J/kg}\cdot\text{K}$

Find: Density at States 1 & 2

$\rightarrow P = \rho R T$ for ideal gases⁺³

+4 for P_1 plot

+4 for P_2 plot

$\rho = \frac{P}{RT}$ $\rightarrow$ calculation in spreadsheet.

Trend in ρ_1 : ρ_1 tracks really closely w/ T_1 . As it gets hotter as the sun comes up, density goes down because pressure is relatively constant.⁺²

2] Given: $\text{CO}_2 \rightarrow M = 44.01 \frac{\text{kg}}{\text{kmol}}$

Assume: ideal gas for Videal

S.C.S. for everything else

Find: $V_{\text{ideal}}, V_{\text{real}}, Z$, phase

where $\rightarrow$ ideal gas: $Pv = RT$ ⁺¹ $\rightarrow v = \frac{RT}{P}$ and $R = \frac{R_u}{M} = \frac{8314.5}{44.01} = 188.92 \frac{\text{J}}{\text{kg}\cdot\text{K}}$ ⁺¹

compressibility factor: $Z = \frac{Pv}{RT}$ ⁺¹

a) $T = 300 \text{ K}$
 $P = 0.2 \times 10^6 \text{ Pa}$

$\rightarrow v = \frac{RT}{P} = \frac{(188.92)(300)}{0.2 \times 10^6}$

$\rightarrow Z = \frac{Pv}{RT} = \frac{(0.2 \times 10^6)(0.28061)}{(188.9)(300)}$

$V_{\text{ideal}} = 0.28338 \frac{\text{m}^3}{\text{kg}}$ ⁺¹
 $V_{\text{real}} = 0.28061 \frac{\text{m}^3}{\text{kg}}$ ⁺¹
 $Z = 0.99$ ⁺¹
 Phase = vapor⁺¹

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b) $T = 300 \text{ K}$
 $P = 4.1 \times 10^6 \text{ Pa}$

$\rightarrow V = \frac{RT}{P} \Rightarrow$

$\rightarrow Z = \frac{Pv}{RT} \Rightarrow$

$V_{\text{ideal}} = 0.0138 \text{ m}^3/\text{kg} +1$
 $V_{\text{NIST}} = 0.010507 \text{ m}^3/\text{kg} +1$
 $Z = 0.76 +1$
 $\text{Phase} = \text{vapor} +1$

c) $T = 300 \text{ K}$
 $P = 8.3 \times 10^6 \text{ Pa}$

$\rightarrow V = \frac{RT}{P} \Rightarrow$

$\rightarrow Z = \frac{Pv}{RT} \Rightarrow$

$V_{\text{ideal}} = 0.0068 \text{ m}^3/\text{kg} +1$
 $V_{\text{NIST}} = 0.0013113 \text{ m}^3/\text{kg} +1$
 $Z = 0.192 +1$
 $\text{Phase} = \text{liquid} +1$

d) $T = 500 \text{ K}$
 $P = 8.3 \times 10^6 \text{ Pa}$

$\rightarrow V = \frac{RT}{P} \Rightarrow$

$\rightarrow Z = \frac{Pv}{RT}$

$V_{\text{ideal}} = 0.01138 \text{ m}^3/\text{kg} +1$
 $V_{\text{NIST}} = 0.010761 \text{ m}^3/\text{kg} +1$
 $Z = 0.9456 +1$
 $\text{Phase} = \text{supercritical} +1$

[3] Assume: S.C.S.

a) $T = 440 \text{ K}$, $P = 0.733679 \text{ MPa}$ $\Rightarrow 0.26090 \text{ m}^3/\text{kg}$

$\rightarrow \text{Table D.1} +1/2$

$\rightarrow \text{Saturated vapor because } V = V_g \text{ for } PrT +1$

$\rightarrow h = h_g = 2764.7 \text{ kJ/kg} +1$

b) $T = 440 \text{ K}, P = 5 \text{ MPa} \rightarrow P > P_{\text{sat}} \rightarrow \text{liquid} +1$

$\rightarrow \text{Table D.4A} +1/2$

$\rightarrow u = 702.2 \text{ kJ/kg} +1$

c) $T = 440 \text{ K}, P = 0.1 \text{ MPa} \rightarrow P < P_{\text{sat}} \rightarrow \text{vapor} +1$

$\rightarrow \text{Table D.3D} +1/2$

$\rightarrow v = 2.0165 \text{ m}^3/\text{kg} +1$

d) $T = 440 \text{ K}, P = 0.733679 \text{ MPa}, v = 0.2 \text{ m}^3/\text{kg}$

$\rightarrow v_f < v < v_g \Rightarrow \text{mixture} +1$

$\rightarrow \text{Table D.1} +1/2$

$\rightarrow h = x h_g + (1-x) h_f \rightarrow x = \frac{v - v_f}{v_g - v_f} = \frac{0.2 - 0.001103}{0.2609 - 0.001103}$

$x = 0.7656 +1$

$h = (0.7656)(2764.7) + (1 - 0.7656)(705.31)$

$h = 2281.98 \text{ kJ/kg} +1$

e) $T = 647.096 \text{ K}, P = 22.064 \text{ MPa}$

$\rightarrow P = P_c, T = T_c \rightarrow \text{critical point} +1$

$\rightarrow \text{Table D.1 or D.2} +1/2$

$\rightarrow v = v_c = 0.0031056 \text{ m}^3/\text{kg} +1$

f) $T = 700 \text{ K}$, $P = 32 \text{ MPa} \rightarrow P > P_c$, $T > T_c \rightarrow \text{supercritical} +1$ $4/7$
 $\rightarrow \text{Table D.3X} +1/2$
 $\rightarrow \rho = 217.87 \text{ kg/m}^3 +1$

g) $T = 605 \text{ K}$, $P = 6.0 \text{ MPa} \rightarrow T > T_{\text{crit}} \rightarrow \text{vapor} +1/2$

$\rightarrow \text{Table D.3N} +1/2$

$\rightarrow \text{need to interpolate} \rightarrow T_a = 600 \text{ K} \quad u_a = 2737.6 \text{ kJ/kg}$

$T_b = 620 \text{ K} \quad u_b = 2783.5 \text{ kJ/kg}$

$T_c = 605 \text{ K}$

$$\Rightarrow u_c = \left(\frac{T_c - T_a}{T_b - T_a} \right) (u_b - u_a) + u_a +1$$

$$= \left(\frac{605 - 600}{620 - 600} \right) (2783.5 - 2737.6) + 2737.6$$

$u_c = 2749.075 \text{ kJ/kg} +1$

Given: $M = 2 \text{ kg}$
state points (below)

Assume: S.C.S. +1
: quasi-g. processes +1

State 1: $P_1 = 5 \text{ MPa}$

$$v_1 = 0.0012864 \text{ m}^3/\text{kg}$$

→ $v = v_f$ on Table D.1 +1

→ saturated liquid

$$\boxed{T_1 = T_{\text{sat}} = 537.09 \text{ K}} +1$$

$$\rightarrow u_1 = u_f = 1148.2 \text{ kJ/kg} +1$$

State 2: $P_2 = 10 \text{ MPa}$

$$T_2 = 560 \text{ K}$$

→ $T_2 < T_{\text{sat}} \rightarrow$ liquid from Table D.2 +1

→ Table D.4B → $u_2 = 1257.6 \text{ kJ/kg} +1$

$$\boxed{v_2 = 0.0013459 \text{ m}^3/\text{kg}} +1$$

State 3: $P_3 = 10 \text{ MPa}$

$$v_3 = 0.020091 \text{ m}^3/\text{kg}$$

→ $v_2 > v_g$ from Table D.2 → vapor +1

→ Table D.3P

$$\boxed{T_3 = 600 \text{ K}} +1$$

$$\rightarrow u_3 = 2619.1 \text{ kJ/kg} +1$$

State 4: $P_4 = 5 \text{ MPa}$

$$v_4 = 0.039446 \text{ m}^3/\text{kg}$$

→ $v_4 = v_g$ at $P_4 \rightarrow$ saturated vapor +1

$$\boxed{T_4 = T_{\text{sat}} = 537.09 \text{ K}} +1$$

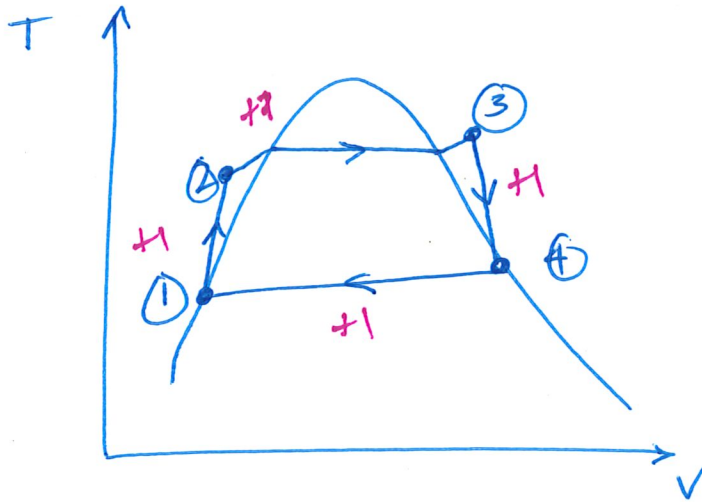
$$\rightarrow u_4 = u_g = 2597.0 \text{ kJ/kg} +1$$

$$\rightarrow U_2 - U_1 = M(u_2 - u_1) = 2(1257.6 - 1148.2) = 218.8 \text{ kJ}^{+1/2} \quad 6/7$$

$$U_3 - U_2 = M(u_3 - u_2) = 2(2619.1 - 1257.6) = 2723 \text{ kJ}^{+1/2}$$

$$U_4 - U_3 = M(u_4 - u_3) = 2(2597 - 2619.1) = -44.2 \text{ kJ}^{+1/2}$$

$$U_1 - U_4 = M(u_1 - u_4) = 2(1148.2 - 2597) = -2897.6 \text{ kJ}^{+1/2}$$



[5] Given: $T_1 = T_2 = 647.096 \text{ K}$
 $P_1 = P_2 = 22.064 \text{ MPa}$
 $T_2 = 500 \text{ K}$

Assume: S.C.S. $+1$
 : quasi-ef. $+1$
 : rigid $\rightarrow v_2 = v_1$ $+1$

Find: p_1 , u_2 , diagram

State 1: $p_1 = p_2 = \frac{1}{v_c} = \frac{1}{0.0031056} \rightarrow \boxed{p_1 = 321.999 \text{ kPa/m}^3}^{+1}$

State 2: $v_2 = v_1 = v_c = 0.0031056 \text{ m}^3/\text{kg}^{+1}$

@ $T_2 = 500 \text{ K} \rightarrow v_f < v_2 < v_g \rightarrow u_2 = x u_g + (1-x) u_f$ $+2$

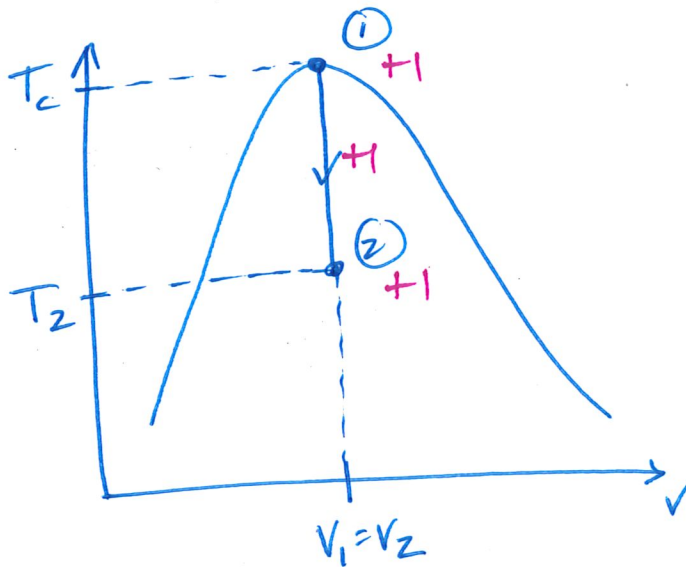
$$x_2 = \frac{v - v_f}{v_g - v_f}^{+2}$$

$$x_2 = \frac{V - V_f}{V_g - V_f} = \frac{0.0031056 - 0.0012029}{0.075764 - 0.0012029} = 0.0255 +1$$

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$$u_2 = (0.0255)(2602.5) + (1 - 0.0255)(972.26)$$

$$u_2 = 1013.83 \text{ kJ/kg} +1$$



[6] Given: $x_1 = 0$ $x_2 = 1$
 $T_1 = 438.1 \text{ K}$ $T_3 = 700 \text{ K}$
 $P_2 = P_1 = P_3$

Assume: S.C.S. +1
 : quasi-eq. +1

Find: h_1, h_2, h_3

State 1: $T_1 = 438.1 \text{ K}, x_1 = 0 \rightarrow h_1 = h_f = 697 \text{ kJ/kg}$ on Table D.2 +1

State 2: $P_2 = P_1 = 0.7 \text{ MPa}, x_2 = 1 \rightarrow h_2 = h_g = 2762.8 \text{ kJ/kg}$ Table D.2 +1

State 3: $P_3 = 0.7 \text{ MPa}, T_3 > T_2 \rightarrow \text{vapor} +1$
 $\rightarrow h_3 = 3326 \text{ kJ/kg}$ Table D.3.H +1